WEEK 2

Supervised Machine Learning Models

1. Objective

This project trains and compares supervised classification and regression models with Scikit-learn using the Titanic passenger dataset. Classification predicts survival; regression predicts fare. A fixed random seed and held-out 20% test set make comparisons repeatable.

2. Dataset and preparation

The source data contains 891 passengers. Predictors are age, sibsp, parch, pclass, sex, and embarked. Missing numeric values are median-imputed and standardized; missing categorical values are mode-imputed and one-hot encoded. Preprocessing is fitted within each pipeline on training data only. The classification split is stratified, and no target-derived feature such as alive is used.

3. Classification results

A most-frequent baseline and four classifiers were evaluated on the same held-out test rows: Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors. Accuracy is overall correctness; precision and recall describe positive-class predictions; F1 balances precision and recall; ROC AUC measures ranking across thresholds.

model	test_rows	accuracy	precision	recall	f1	roc_auc
Logistic Regression	179	0.804	0.793	0.667	0.724	0.844
Decision Tree	179	0.793	0.833	0.580	0.684	0.820
K-Nearest Neighbors	179	0.788	0.816	0.580	0.678	0.834
Random Forest	179	0.771	0.750	0.609	0.672	0.850
Most-frequent baseline	179	0.615	0.000	0.000	0.000	0.500

4. Regression results

Linear Regression is compared with a mean-fare Dummy Regressor baseline. MAE is average absolute error; RMSE penalizes large errors more strongly; R^2 compares the model against a mean prediction and can be negative.

model	test_rows	mae	rmse	r2
Linear Regression	179	18.870	30.920	0.382
Mean fare baseline	179	25.692	39.383	-0.002

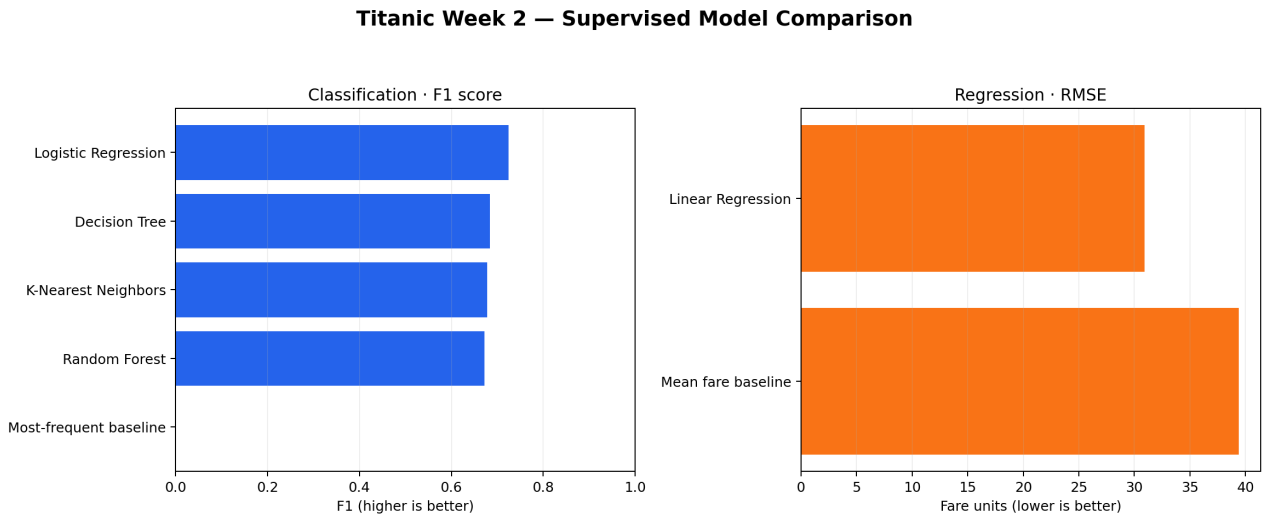


Figure 1. F1 score and RMSE comparisons on held-out test sets; higher F1 and lower RMSE are preferred.

5. Conclusion and limitations

The experiment demonstrates model training, prediction, and metric-based comparison. Results are specific to one holdout split, are not causal findings, and should not be treated as production performance. Cross-validation, tuning, and analysis of the skewed fare target are useful follow-up steps.

Repository: https://github.com/shreyas23dev/Week1